

Employee Success Analytics at NextGen Corp.

SQL CAPSTONE PRESENTATION

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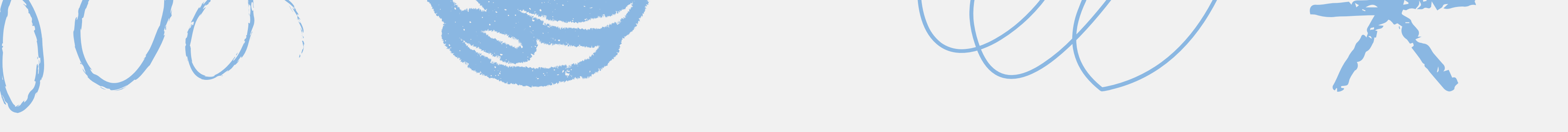
BUSINESS OVERVIEW

NextGen Corp. is a growing technology company focused on developing innovative solutions in the software and hardware spaces. The company prides itself on attracting top talent and maintaining high employee satisfaction to drive growth. However, there are increasing concerns regarding employee turnover, performance variability, and salary disparities within departments.

To ensure continued success, NextGen Corp. needs to optimize employee retention, track employee performance consistently, and maintain fair salary structures across departments. The HR department needs a data-driven approach to:

- Identify trends and patterns in employee retention and turnover.
- Track and evaluate performance across different departments.
- Assess the relationship between salary and performance to ensure fairness and employee satisfaction.





ANALYSIS FLOW



01

**Employee Retention
Analysis**

02

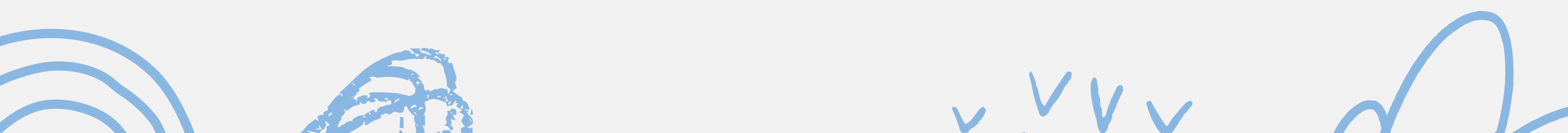
**Performance
Analysis**

03

**Salary
Analysis**

04

Recommendations





EMPLOYEE INSIGHT

Understand the employee turnover trends and
identify the root causes of high turnover
rates.

Who are the top 5 highest serving employees?

```
SELECT e.employee_id, concat(e.first_name, ' ', e.last_name)
AS Full_name, e.hire_date
FROM employee e
ORDER BY "hire_date" ASC
LIMIT 5;
```

	employee_id [PK] integer	full_name text	hire_date date
1	8	David Moore	2015-06-30
2	48	Frank Smith	2015-09-08
3	67	Jane Brown	2015-10-12
4	44	John Johnson	2015-10-27
5	57	John Doe	2016-02-16

The top 5 longest-serving employees have each been with the organisation for over 9 years (3,449–3,680 days).

- David Moore has the longest tenure at 3,680 days (~10 years).
- All five employees were hired between June 2015 and February 2016, showing stability in that hiring cohort.

This indicates strong employee retention for hires made during that period.

What is the turnover rate for each department?

```
SELECT d.department_name, ROUND(CAST(COUNT(t.turnover_id) AS DEC  
COUNT(e.employee_id) * 100, 2) AS turnover_rate_percent  
FROM department d  
JOIN employee e ON d.department_id = e.department_id  
LEFT JOIN turnover t ON e.employee_id = t.employee_id  
GROUP BY d.department_name  
ORDER BY turnover_rate_percent DESC;
```

	department_name character varying (30) 🔒	turnover_rate_percent numeric 🔒
1	Marketing	92.86
2	Engineering	66.67
3	Sales	27.59
4	HR	27.27

- **Marketing** has an extremely high turnover rate at 92.86%, followed by **Engineering** at 66.67% which suggests instability and possible issues with employee retention in these departments.
- **Sales** and **HR** have much lower turnover rates, both below 30%, indicating more stability.

Which employees are at risk of leaving based on their performance?

Grace Wilson from the Sales department has been identified as at risk of leaving based on her low average performance score of 3.43.

```
SELECT e.employee_id,
       CONCAT(e.first_name, ' ', e.last_name) AS full_name,
       d.department_name,
       COUNT(p.performance_id) AS total_reviews,
       ROUND(AVG(p.performance_score), 2) AS avg_performance_score,
       CASE
         WHEN AVG(p.performance_score) <= 3.0 THEN 'High Risk'
         WHEN AVG(p.performance_score) <= 3.5 THEN 'Moderate Risk'
         ELSE 'Low Risk'
       END AS risk_level
FROM employee e
JOIN performance p ON p.employee_id = e.employee_id
JOIN department d ON e.department_id = d.department_id
GROUP BY e.employee_id, full_name, d.department_name
HAVING AVG(p.performance_score) <= 3.5
ORDER BY avg_performance_score;
```

id	full_name	department_name	total_reviews	avg_performance_score	risk_level
	text	character varying (30)	bigint	numeric	text
22	Grace Wilson	Sales	6	3.43	Moderate Risk

What are the main reasons employees are leaving the company?

```
SELECT
  t.reason_for_leaving,
  COUNT(*) AS number_of_exits,
  ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS percentage_of_total_exits
FROM turnover t
JOIN employee e ON t.employee_id = e.employee_id
GROUP BY t.reason_for_leaving
ORDER BY number_of_exits DESC;
```

reason_for_leaving 	number_of_exits 	percentage_of_total_exits 
text	bigint	numeric
Personal	11	39.29
Found Another Job	7	25.00
Career Growth	5	17.86
Retired	5	17.86

The most common reason employees leave is Personal reasons, accounting for 39.29% of total exits (11 employees), followed by Finding another job at 25.00% (7 employees). 42.86% of exits (Found Another Job + Career Growth) indicate that employees are leaving for external opportunities, which could signal unmet development or career satisfaction internally.



PERFORMANCE INSIGHT

Evaluate employee performance across different departments and identify areas where performance can be improved.

How many employees has left the company?

```
SELECT COUNT(*) AS employees_left  
FROM turnover;
```

	employees_left bigint
1	28

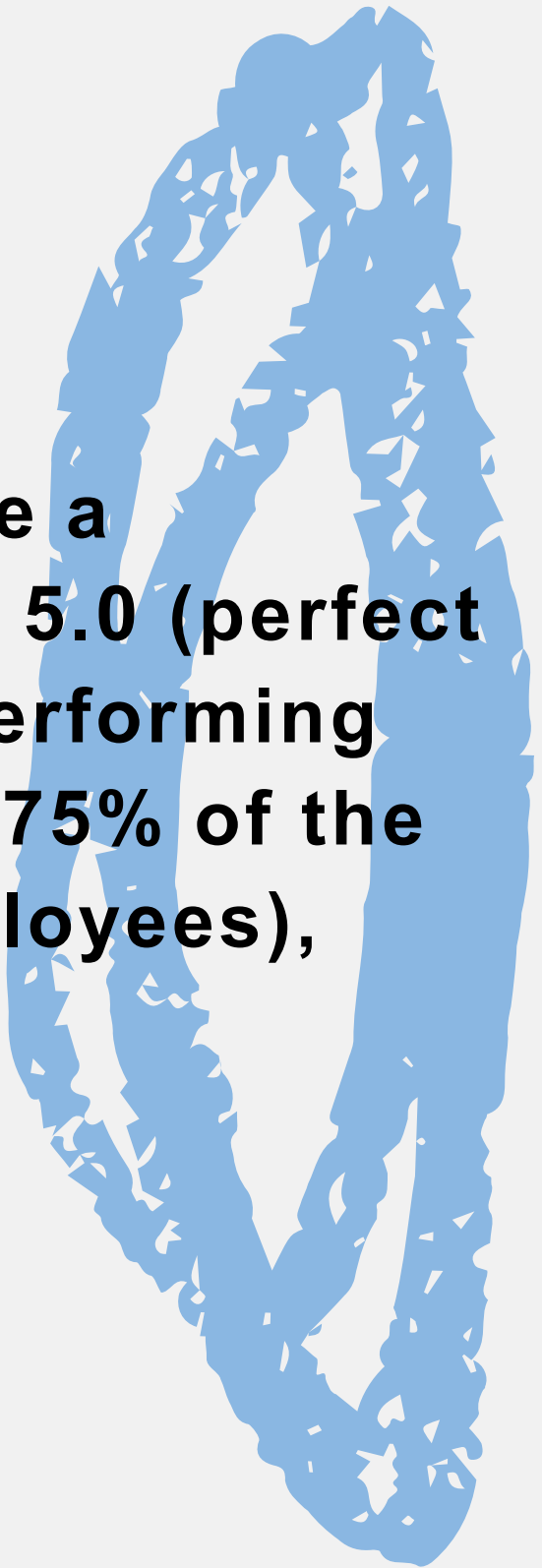
Nearly half of the workforce, a total of 28 employees have left the company out of 60. This is a high exit rate and may indicate underlying issues with employee satisfaction, engagement, or retention strategies.

How many employees have a performance score of 5.0 / below 3.5?

```
SELECT COUNT(DISTINCT employee_id) as employee_count
FROM performance
WHERE performance_score = 5 or performance_score<3.5;
```

	employee_count bigint
1	47

A total of 47 employees have a performance score of either 5.0 (perfect score) or below 3.5 (underperforming range). It accounts for over 75% of the workforce (47 out of 60 employees), which is unusually high.

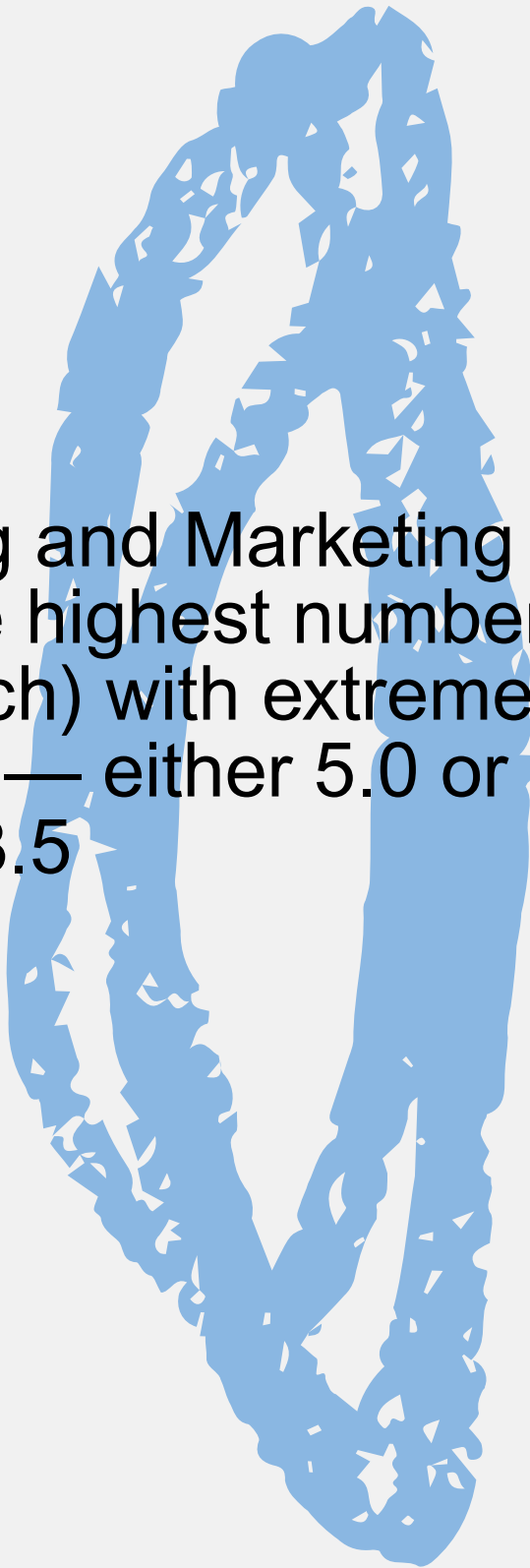


Which department has the most employees with a performance of 5.0 / below 3.5?

```
SELECT d.department_name, COUNT(DISTINCT p.employee_id) AS employee_count
FROM performance p
JOIN department d ON p.department_id=d.department_id
WHERE p.performance_score = 5.0 OR p.performance_score<3.5
GROUP BY d.department_name
ORDER BY employee_count DESC
LIMIT 3;
```

	department_name character varying (30) 🔒	employee_count bigint 🔒
1	Engineering	17
2	Marketing	17
3	Sales	7

Both the Engineering and Marketing departments have the highest number of employees (17 each) with extreme performance scores — either 5.0 or below 3.5



What is the average performance score by department?

```
SELECT d.department_name, ROUND(AVG(p.performance_score), 2) AS avg_performance_
FROM employee e
JOIN performance p ON e.employee_id = p.employee_id
JOIN department d ON e.department_id = d.department_id
GROUP BY d.department_name
ORDER BY avg_performance_score DESC;
```

	department_name character varying (30) 🔒	avg_performance_score numeric 🔒
1	Marketing	4.17
2	HR	4.11
3	Sales	4.07
4	Engineering	4.01

Both the Engineering and Marketing departments have the highest average performance score which shows a high concentration of performance extremes



SALARY INSIGHT

Analyze salary distribution and ensure fair compensation based on performance and departmental benchmarks.

What is the total salary expense for the company?

```
SELECT COUNT(employee_id) AS no_of_employees, TO_CHAR(SUM(salary_amount):: numeric, 'FM£999,999,999.00')
AS total_salary_expense
FROM salary;
```

	no_of_employees bigint	total_salary_expense text
1	60	£4,850,000.00

The total salary expense for the company is £4,850,000.00 for 60 employees.



What is the average salary by job title?

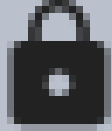
```
SELECT e.job_title, ROUND(AVG(s.salary_amount), 2) AS avg_salary
FROM employee e
LEFT JOIN salary s ON s.employee_id = e.employee_id
GROUP BY job_title
ORDER BY avg_salary DESC
```

	job_title character varying (30) 🔒	avg_salary numeric 🔒
1	Sales Representative	84285.71
2	HR Specialist	81818.18
3	Engineer	80000.00
4	Sales Manager	80000.00
5	Marketing Specialist	77857.14

- Sales Representatives have the highest average salary at £84,285.71, even higher than Sales Managers.
- Marketing Specialists have the lowest average salary among the listed roles, at £77,857.14, despite Marketing being one of the departments with a high number of top performers.
- The salary spread is relatively narrow (within ~£6,400), indicating a fairly balanced compensation structure, though some roles may warrant review for alignment with performance or responsibility levels.

How many employees earn above 80,000?

```
SELECT COUNT(DISTINCT s.employee_id) AS high_earners
FROM salary s
WHERE s.salary_amount > 80000;
```

	high_earners bigint 
1	26

26 employees earn above £80,000 annually.

A significant portion of staff are in higher salary brackets, suggesting the company places strong emphasis on hiring or retaining specialised or high-value roles, likely in areas like Sales, Engineering, or HR.

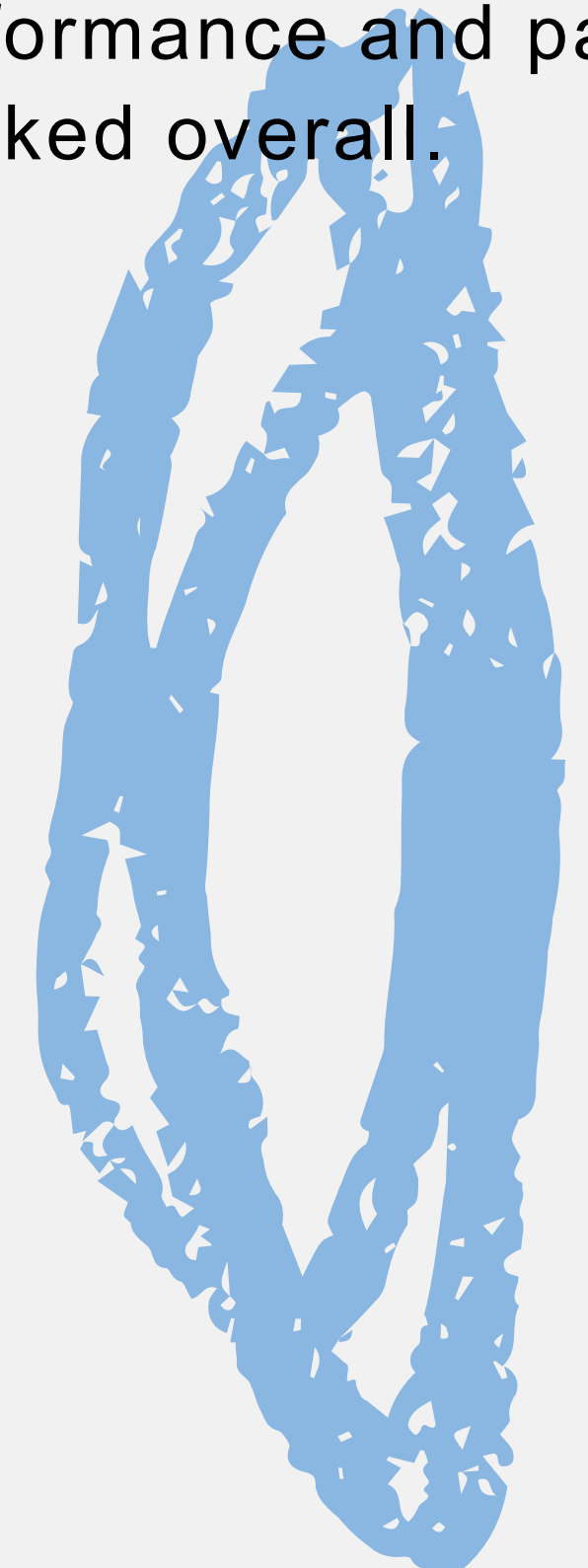


How does performance correlate with salary across departments?

```
SELECT
  d.department_name,
  ROUND(AVG(p.performance_score),2) AS avg_perf_score,
  ROUND(AVG(s.salary_amount),2) AS avg_salary,
  ROUND(CORR(s.salary_amount,p.performance_score)::numeric,2)
  AS perf_salary_correllation
FROM employee e
JOIN performance p ON e.employee_id = p.employee_id
JOIN salary s ON e.employee_id = s.employee_id
JOIN department d ON e.department_id = d.department_id
GROUP BY d.department_name;
```

department_name character varying (30) 🔒	avg_perf_score numeric 🔒	avg_salary numeric 🔒	perf_salary_correllation numeric 🔒
Engineering	4.01	80000.00	0.21
HR	4.11	81818.18	0.16
Sales	4.07	82068.97	0.10
Marketing	4.17	77857.14	-0.07

The correlations across all departments are relatively weak, indicating that performance and pay are not strongly linked overall.



Recommendations

Leadership & Mentorship Roles:

Leverage their experience by assigning them to mentorship programs, onboarding new employees, or involving them in decision-making processes.

Recognition Programs:

Consider formally recognising their loyalty and contributions through awards, bonuses, or internal communications to boost morale and encourage retention across the wider team.

Conduct a One-on-One Check-In:

Explore potential performance blockers such as workload, motivation, support, or external factors. Reinforce her value to the team while actively listening to her concerns.



Performance Development:

For employees scoring <3.6 , consider coaching, skill development, or feedback sessions. Introduce personal development plans tied to quarterly goals and peer reviews.

Conduct performance reviews to identify skill gaps, unclear expectations, or external factors impacting results.

Strengthen Career Development:

Create visible career progression pathways and upskilling opportunities to reduce exits from career-driven employees. Encourage internal mobility before employees seek external roles

Deep-Dive Review for Both Departments:

Explore if roles, responsibilities, or expectations are misaligned, leading to this performance gap. Conduct team-level analysis to identify whether underperformance is widespread or isolated. Consider implementing or refining performance-based incentives or bonuses

The background is a light gray color, decorated with various hand-drawn blue doodles. These include several loops and swirls at the top, a series of vertical lines on the right side, and a wavy line and several checkmarks at the bottom. The central text is in a bold, black, sans-serif font with a white drop shadow.

**Thank you
very much!**